

under the directory `/etc/logrotate.d/`. We have defined `/etc/logrotate.d/jenkins` especially for our example to take care of Jenkins server logs.

A sample `config` file defined to rotate a `jenkins.log` that is 4MB in size is as follows:

```
[root@ci-01 ~]# cat /etc/logrotate.d/jenkins
/var/log/jenkins/jenkins.log {
compress
maxage 60
rotate 99
size=+4096k
notifempty
missingok
create 644
copytruncate
}
```

To learn more about each parameter defined here and to check for additional options, you can refer to <https://linux.die.net/man/8/logrotate>.

Adding logrotate to cron

For logrotate to run daily, it must be added under a daily anacron, i.e., `/etc/cron.daily/logrotate`, using the code given below (this is different from regular cron):

```
#!/bin/sh
/usr/sbin/logrotate /etc/logrotate.conf
EXITVALUE=$?
if [ $EXITVALUE != 0 ]; then
/usr/bin/logger -t logrotate "ALERT exited abnormally
with [$EXITVALUE]"
fi
exit $EXITVALUE
```

If you want logrotate to run hourly, just copy the logrotate script from the previously defined `/etc/cron.daily/logrotate` or create a new script under `/etc/cron.hourly/logrotate` with the above script.

In case you want to run logrotate more often, like every few minutes or so, then you can add it to regular cron (`crontab -e`). The code given below runs logrotate every five minutes:

```
* /5 * * * * /etc/cron.hourly/logrotate
```

Points to note

1. Log files that are rotated daily will have similar names like `logfile-name-date` (*Ex-jenkins-new.log-20220701.gz*), since we defined the naming format as `'dateext'` in `/etc/logrotate.conf` as shown below:

```
# use date as a suffix of the rotated file
dateext
```

If the frequency to rotate logs is changed to hourly or after every few minutes, then multiple files are rotated within the same day if log files grow. This can cause duplicates in the names of the rotating log files. To avoid this, add `%Y%m%d-%s` to the format as listed below:

```
# Use date as a suffix of the rotated file
dateformat -%Y%m%d-%s
dateext
```

2. To run logrotate manually, just execute `/etc/cron.hourly/logrotate`.
3. In Jenkins, files sometimes grow exponentially. For example, the `jenkins.log` can grow beyond a few GBs in size within hours or even in minutes in some rare cases, and can occupy the whole disk space, which can cause Jenkins to stall.

In that situation, deleting `jenkins.log` will not free up the disk space immediately as we may expect it to do, since some processes will still be using the log. These can be found by checking with `lsof`:

```
[jenkins@jenkins01 jenkins]$ lsof +L1
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NLINK NODE NAME
java 1420 jenkins 14w REG 253,0 14706807770 0 67837765 /
var/log/jenkins/jenkins-new.log (deleted)
```

Killing the processes will disturb the running of Jenkins. You need to restart Jenkins so that these process IDs are released, and the system can claim free space by deleting the overgrown `jenkins.log` or problematic log file.

Testing log rotation

To test whether the log rotation is functioning correctly based on the log file size, you can use the command `urandom` to populate log files instantly, and see whether logs are rotated or not when logrotate is called by the cron or manually. The example given below populates the log file by 10MB:

```
sudo base64 /dev/urandom | head -c 10000000 >/var/log/
jenkins/jenkins-new.log
```

END 

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